Sanjana Shivanand

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EDUCATION

Binghamton University, SUNY | Binghamton, NY *Jan 2025 – Dec 2026*
Master of Science in Computer Science

JSS Academy of Technical Education | Bangalore, India
Bachelor of Engineering in Computer Science *June 2024*

TECHNICAL SKILLS

Programming and Fundamentals: Python, C++, SQL, JavaScript, TypeScript, OOP, Data Structures and Algorithms
Full-Stack Development: React, Next.js, FastAPI, Django, REST APIs, ApexCharts
Databases and Cloud: PostgreSQL, MariaDB, DuckDB, AWS (S3, Athena, Lambda), Azure (ADF, ADLS Gen2, SQL)
Development tools and Automation: Docker, Git, GitHub Actions, CI/CD, pytest, Power Apps, Power Automate
AI & Data Engineering: LangChain, LangGraph, RAG, scikit-learn, XGBoost, PySpark, Airflow, Prefect, dbt

WORK EXPERIENCE

Metropolitan Transportation Authority | Staten Island, USA *Jan 2026 – Present*
Software Development Intern – Operations Support

- Implemented a React-based transit analytics platform using public MTA data, displaying pre-aggregated ridership and disruption insights to reduce repeated database queries by 40% and lower compute costs.
- Maintained Python data ingestion and validation modules with Azure Data Factory, using ADLS Gen2 and Azure SQL to prepare MTA datasets for station-level ridership trends and service-delay cause analysis.
- Integrated Power Automate with SharePoint and internal REST APIs to automate Staten Island Policy & Compliance reporting for disciplinary records of 600+ employees, reducing report preparation time by 72%.

Centre for Development of Telematics | Bangalore, India *Sept 2023 – Oct 2024*
Software Development Intern – Network Management

- Developed React components and Django HTTP endpoints for a 5G O-RAN analytics portal, enabling dataset uploads, merging, and ML result visualization through a unified interface.
- Applied session-based authentication, role-based authorization, and activity logging using Django, password hashing, and MariaDB via Django ORM to block unauthorized admin access across user roles.
- Trained regression and LSTM models using scikit-learn, XGBoost, and PyTorch for 5G Network slicing and V2X Communications, by automating Django dashboards comparisons to cut estimated evaluation time by 67%.

PROJECTS

SafeEats NYC - Restaurant Safety & Prediction System | *GitHub*

- Developed FastAPI services and a React dashboard to search NYC restaurant inspection histories and display B/C-grade risk estimates for 24K+ eligible restaurants, using precomputed Athena/S3 snapshots on Vercel.
- Built Airflow ETL pipelines using PySpark and idempotent S3 outputs to process 398K cleaned records, remove 3K duplicates, and validate data with 119 dbt tests while automating code checks through GitHub Actions.

FinSight AI - Financial Intelligence Platform | *GitHub*

- Created a Dockerized FastAPI with Streamlit platform to analyze prices, volatility, and news sentiment across 25 assets, using 3-hour caching and stored-data fallbacks to handle API failures.
- Integrated an MCP server and LangGraph agent to generate financial reports using Qdrant-indexed SEC filings, and orchestrated Prefect pipeline validated by 21 dbt tests.

Social Media Intelligence - Discussion Trends & Monitoring Platform | *GitHub*

- Connected a FastAPI backend to analyze engagement and toxicity across 3.6M+ Reddit and 4chan posts in two PostgreSQL databases, integrating a LangGraph agent to convert natural-language questions into SQL queries.
- Designed React dashboard views with ApexCharts and date/community filters to explore posting and event trends, using TanStack Query to reuse cached API data.